Table of Contents

2012 2014

1.

2.

3.

RFM Recency
Frequency Monetary RFM

“ M” “
C”
L
L R F M C
LRFMC LRFMC

LRFMC K-Means

1

2 1

3 2 LRFMC

4

- pandas K-means LRFCM

1.

2014-03-31
62 988

44

0 0 0
0

In [1]:
```
import pandas as pd

datafile= './data/air_data.csv' #          ,
resultfile = './tmp/explore.xlsx' #

data = pd.read_csv(datafile, encoding = 'utf-8') #            UTF-8

explore = data.describe(percentiles = [], include = 'all').T #
explore['null'] = len(data)-explore['count'] #describe()
explore = explore[['null', 'max', 'min']]
explore.columns = [u'      ', u'      ', u'      '] #

'''
```

```
describe()                           count             unique              top
explore.to_excel(resultfile) #
```

In [2]:

```
explore
```

Loading [MathJax]/extensions/Safe.js

Out[2]:

MEMBER_NO	0.0	62988.0	1.0
FFP_DATE	0	NaN	NaN
FIRST_FLIGHT_DATE	0	NaN	NaN
GENDER	3	NaN	NaN
FFP_TIER	0.0	6.0	4.0
WORK_CITY	2269	NaN	NaN
WORK_PROVINCE	3248	NaN	NaN
WORK_COUNTRY	26	NaN	NaN
AGE	420.0	110.0	6.0
LOAD_TIME	0	NaN	NaN
FLIGHT_COUNT	0.0	213.0	2.0
BP_SUM	0.0	505308.0	0.0
EP_SUM_YR_1	0.0	0.0	0.0
EP_SUM_YR_2	0.0	74460.0	0.0
SUM_YR_1	551.0	239560.0	0.0
SUM_YR_2	138.0	234188.0	0.0
SEG_KM_SUM	0.0	580717.0	368.0
WEIGHTED_SEG_KM	0.0	558440.14	0.0
LAST_FLIGHT_DATE	0	NaN	NaN
AVG_FLIGHT_COUNT	0.0	26.625	0.25
AVG_BP_SUM	0.0	63163.5	0.0
BEGIN_TO_FIRST	0.0	729.0	0.0
LAST_TO_END	0.0	731.0	1.0
AVG_INTERVAL	0.0	728.0	0.0
MAX_INTERVAL	0.0	728.0	0.0
ADD_POINTS_SUM_YR_1	0.0	600000.0	0.0
ADD_POINTS_SUM_YR_2	0.0	728282.0	0.0
EXCHANGE_COUNT	0.0	46.0	0.0
avg_discount	0.0	1.5	0.0
P1Y_Flight_Count	0.0	118.0	0.0
L1Y_Flight_Count	0.0	111.0	0.0
P1Y_BP_SUM	0.0	246197.0	0.0
L1Y_BP_SUM	0.0	259111.0	0.0
EP_SUM	0.0	74460.0	0.0
ADD_Point_SUM	0.0	984938.0	0.0
Eli_Add_Point_Sum	0.0	984938.0	0.0
L1Y_ELi_Add_Points	0.0	728282.0	0.0

Points_Sum	0.0	985572.0	0.0
L1Y_Points_Sum	0.0	728282.0	0.0
Ration_L1Y_Flight_Count	0.0	1.0	0.0
Ration_P1Y_Flight_Count	0.0	1.0	0.0
Ration_P1Y_BPS	0.0	0.999989	0.0
Ration_L1Y_BPS	0.0	0.999993	0.0
Point_NotFlight	0.0	140.0	0.0

2.

LRMFC
44

0
0 0
1.
2. 0 0 0 pandas

In [3]:
```
#

import pandas as pd

datafile= './data/air_data.csv' #           ,
cleanedfile = './tmp/data_cleaned.csv' #

data = pd.read_csv(datafile,encoding='utf-8') #              UTF-8

#
data = data[data['SUM_YR_1'].notnull()&data['SUM_YR_2'].notnull()]

#                                          0
index1 = data['SUM_YR_1'] != 0
index2 = data['SUM_YR_2'] != 0
index3 = (data['SEG_KM_SUM'] == 0) & (data['avg_discount'] == 0) #
data = data[index1 | index2 | index3] #        “  ”
data.head()
```

Out[3]:

	MEMBER_NO	FFP_DATE	FIRST_FLIGHT_DATE	GENDER	FFP_TIER	WORK_CITY	WOF
0	54993	2006/11/02	2008/12/24		6	.	
1	28065	2007/02/19	2007/08/03		6	NaN	
2	55106	2007/02/01	2007/08/30		6	.	
3	21189	2008/08/22	2008/08/23		5	Los Angeles	
4	39546	2009/04/10	2009/04/15		6		

5 rows × 44 columns

In [4]:

```
data.to_csv(cleanedfile) #       ,    excel    csv         ansi
```

LRFMC LRMFC

6 FFP_DATE LOAD_TIME FLIGHT_COUNT AVG_DISCOUNT SEG_KM_SUM LAST_TO_END

In [5]:

```
outfile = './tmp/data_stipu.csv'
df = data[['FFP_DATE','LOAD_TIME','avg_discount','FLIGHT_COUNT','SEG_KM_S
df.head()
```

Out[5]:

	FFP_DATE	LOAD_TIME	avg_discount	FLIGHT_COUNT	SEG_KM_SUM	LAST_TO_END
0	2006/11/02	2014/03/31	0.961639	210	580717	1
1	2007/02/19	2014/03/31	1.252314	140	293678	7
2	2007/02/01	2014/03/31	1.254676	135	283712	11
3	2008/08/22	2014/03/31	1.090870	23	281336	97
4	2009/04/10	2014/03/31	0.970658	152	309928	5

In [6]:

```
df.to_csv(outfile)
```

3.

LRFMC

1 L = LOAD_TIME - FFP_DATE =
- [ ]

2 R = LAST_TO_END =
[ ]

3 F = FLIGHT_COUNT =
[ ]

Loading [MathJax]/extensions/Safe.js

4 M = SEG_KM_SUM =
[ ]

5 C = AVG_DISCOUNT =
[ ]

5 5

,

ZL,ZR,ZF,ZM,ZC5

In [7]:
```
datafile = './tmp/data_stipu.csv'
standrandfile = './tmp/data_stand.csv'

data = pd.read_csv(datafile)

#
df = data[['LAST_TO_END','FLIGHT_COUNT','SEG_KM_SUM','avg_discount']].cop
df['L'] = (pd.to_datetime(data['LOAD_TIME']) - pd.to_datetime(data['FFP_D
df.rename(columns={'LAST_TO_END':'R','FLIGHT_COUNT':'F','SEG_KM_SUM':'M',

#
df = (df - df.mean(axis=0))/(df.std(axis=0))#
df.columns = ['z'+i for i in df.columns]#

df.to_csv(standrandfile)
```

In [8]:
```
df
```

Out[8]:

	zR	zF	zM	zC	zL
0	-0.944948	14.034016	26.761154	1.295540	1.435707
1	-0.911894	9.073213	13.126864	2.868176	1.307152
2	-0.889859	8.718869	12.653481	2.880950	1.328381
3	-0.416098	0.781585	12.540622	1.994714	0.658476
4	-0.922912	9.923636	13.898736	1.344335	0.386032
...	...	...	...	...	...
62039	-0.460169	-0.706656	-0.805297	-0.065898	2.076128
62040	-0.283886	-0.706656	-0.805297	-0.282309	0.557046
62041	-0.735611	-0.706656	-0.772332	-2.689885	-0.149421
62042	1.605649	-0.706656	-0.779837	-2.554628	-1.206173
62043	0.603039	-0.706656	-0.786677	-2.392319	-0.479656

62044 rows × 5 columns

4

Loading [MathJax]/extensions/Safe.js

1. LRFMC 5

2.

K-means

5

K-Means sklearn

In [9]:
```python
import pandas as pd
from sklearn.cluster import KMeans #    K
import numpy as np

np.set_printoptions(threshold=np.inf)

inputfile = './tmp/data_stand.csv' #
k = 5                           #

#           5
data = pd.read_csv(inputfile).drop(['Unnamed: 0'],axis=1) #
data.head()
```

Out[9]:

	zR	zF	zM	zC	zL
0	-0.944948	14.034016	26.761154	1.295540	1.435707
1	-0.911894	9.073213	13.126864	2.868176	1.307152
2	-0.889859	8.718869	12.653481	2.880950	1.328381
3	-0.416098	0.781585	12.540622	1.994714	0.658476
4	-0.922912	9.923636	13.898736	1.344335	0.386032

In [10]:
```python
#    k-means
kmodel = KMeans(n_clusters = k)
kmodel.fit(data) #

print(kmodel.cluster_centers_) #
```

```
C:\Users\Administrator\Envs\jv\lib\site-packages\sklearn\cluster\_kmeans.
py:870: FutureWarning: The default value of `n_init` will change from 10
to 'auto' in 1.4. Set the value of `n_init` explicitly to suppress the wa
rning
  warnings.warn(
[[-0.00270016 -0.23214985 -0.23645382  2.17019818  0.0416837 ]
 [-0.41509319 -0.16064594 -0.16035949 -0.25804693 -0.70029848]
 [-0.79940684  2.48313493  2.42423773  0.3097848   0.48354786]
 [-0.37743534 -0.08663489 -0.09454128 -0.15689175  1.16093248]
 [ 1.68695486 -0.57391605 -0.53673989 -0.17521822 -0.31318596]]
```

K-Means

5

In [11]:

```
import numpy as np
import matplotlib.pyplot as plt

labels = data.columns.tolist() #
k = 5 #

plot_data = kmodel.cluster_centers_
color = ['b', 'g', 'r', 'c', 'y'] #

angles = np.linspace(0, 2*np.pi, k, endpoint=False)
plot_data = np.concatenate((plot_data, plot_data[:,[0]]), axis=1) #
angles = np.concatenate((angles, [angles[0]])) #
labels.append('zR')

print(plot_data)
print(labels)
```

```
[[-0.00270016 -0.23214985 -0.23645382  2.17019818  0.0416837  -0.0027001
6]
 [-0.41509319 -0.16064594 -0.16035949 -0.25804693 -0.70029848 -0.4150931
9]
 [-0.79940684  2.48313493  2.42423773  0.3097848   0.48354786 -0.7994068
4]
 [-0.37743534 -0.08663489 -0.09454128 -0.15689175  1.16093248 -0.3774353
4]
 [ 1.68695486 -0.57391605 -0.53673989 -0.17521822 -0.31318596  1.6869548
6]]
['zR', 'zF', 'zM', 'zC', 'zL', 'zR']
```

In [12]:

```
#   5
fig = plt.figure()
ax = fig.add_subplot(111, polar=True) #polar

for i in range(len(plot_data)):
    ax.plot(angles, plot_data[i], 'o-', color = color[i], label = u'custo
ax.set_rgrids(np.arange(0.01, 3.5, 0.5), np.arange(-1, 2.5, 0.5), fontpro
ax.set_thetagrids(angles * 180/np.pi, labels, fontproperties="SimHei")

plt.legend(loc = 4)
plt.show()
```

1 R F M 2 C 3
F M R 4 L R 5 L

LRMFC L\M\F\C R

1		RFM
2	C	RFM
3	FMR	
4		LC
5	LFM	

- C
  R F M

- C R
  F M L

- C F M
  R

- C
  R F M L

1 R FM
2 C R FM
3 F,M R
4 L,C
5 L F,M

In [ ]: